# Resource list

- ant-honesty https://alignment.anthropic.com/2025/honesty-elicitation
- https://www.tensortonic.com/ml-math/information-theory/kl-divergence
- Nnsight-status https://nnsight.net/status/
- nnsight-chat-templates https://nnsight.net/tutorials/tutorials/get_started/chat_templates/#setup

- Liars-bench https://arxiv.org/pdf/2511.16035

- mo-misalignment https://arxiv.org/pdf/2506.11613

- geometry-tegmark https://arxiv.org/pdf/2310.06824 (this is the geometry of truth paper, cite [Tegmark 2024])
- obfuscation-atlas https://arxiv.org/pdf/2602.15515

- Ishaan's jupyter notebook for experiments with NDIF models - https://colab.research.google.com/drive/11b9x88-hmLkrUbgFZrz5Xx2LB1q0ty81#scrollTo=fiNvNH2P2ev0
- [Mahon 2016] - https://plato.stanford.edu/archives/win2016/entries/lying-definition/#NonDec
-

# Submitted Proposal (<2k lines)

We began w/ this hypothesis: if a model organism is induced to lie, maybe it becomes less assistant-like while doing so. Extending that to lie detection - if the model is lying, perhaps its persona has steered away from the assistant persona it holds when honest, and testing for this should be enough. Reason for hypothesis - linear, causally implicated representations of truth emerge with scale (Tegmark 2024)[https://arxiv.org/pdf/2310.06824)

However, this is difficult to generalize. Its assistant persona may itself have been modified, or it may have learned a hidden lying persona orthogonal to assistantness. We faced similar generalization issues w/ other ideas too, and realized an "ensemble" of methods can be more robust and general than individual ones.

Proposal: build a composite lie score gathering the knowledge/output gap (a lie per Liars' Bench) using multiple methods.

First, a Concealed Information Test-style probe - a knowledge detector that provides the LLM options, one of which is the truth (the trusted model can generate these option-enhanced prompts). We train linear probes for each option and look for differences in activations. This can also act as a simpler black box test.

Second, forcing the MO to generate the honest answer from the trusted model, then training probes to find activation differences. For MCQs, forcing output to structured JSON or "<A/B/C/D> is the correct output" should be enough. For natural language, we can use structured JSON or find the least common prefix between honest and dishonest responses.

Finally, our earlier hypothesis: probing for un-assistant-like behaviour.

We also aim to use more robust lie detectors (multilayer ensembling) and check for obfuscated activations (in case MOs are posttrained to fool probes). How we devise the composite score and threshold is open, and will be clearer once we test efficacy on the open dataset. We're also concerned about accidentally generating confounding behaviours in CIT prompting, prefill, and persona probing - we want to keep these in mind so as not to compromise robustness.

lmk your thoughts

# Definitions

**BELIEF**

There is a paper [Ren et al. 2025] on the MASK benchmark which is what Liar's Bench uses to operationalise beliefs through consistent responses in absence of a direct pressure to lie.

**LYING**

According to the contest, a *lie* made by a model is a *statement* that the model itself *believes* to be false. Here, *belief* is established by demonstrating a consistency of *statement* made by the model when the pressure imposed upon it to *lie* is removed.

The paper Liar's Bench categorises lying across two axes:

1. **<u>Object of belief</u>**: The model puts forward an *assertion* (which is a *statement* proposed by the agent about a *proposition* ***P*** with an intent to communicate that it is made with a *truthful* intent, or in other words, the *statement* is made with the *intent* to *warrant* the truth of the statement within the context in which it is made.)
   <u>Note: **P** can be learnt at any time during training or dialogue!</u>
   <u>Note: **P** can be</u>
   a. **<u>World-knowledge</u>** "France is the capital of Paris"
   b. **<u>Self-knowledge</u>**
      i. **<u>Capabilities/properties</u>** "I do not have access to the internet"
      ii. **<u>Past-actions</u>** "I saved your file to disk"
      iii. **<u>Policies</u>** "I will not insert backdoors"

2. **<u>Reason for lying</u>**:
    a. Encouraged: it can be *context-dependent* for example, when the agent is told to believe in **P** or it is told to roleplay a persona that implies belief in **P**, etc.
    b. Inherent: it can be *context-independent* for example, arising from behavioural patterns learnt during post-training, in which case it will be generally held, unless overwritten.

According to the contest, we are to refer strictly to the assertion-based definition operationalised in Liar's bench. Liar's bench itself cites [Mahon 2016] declaring its scope to target the ***<u>non-deceptionist, assertion-based</u>*** definition of lying.

> To ensure the concepts we define are measurable in LLMs, we adopt a *non-deceptionist* definition of lying (Mahon, 2016). This differentiates lying from deception as it does not require deceptive intent, which is particularly challenging to verify in LLMs.
>
> Lying is therefore defined in our work using an assertion-based definition similar to Fallis (2009): **the assistant *lies* if it states something it believes to be false.** For example, if it generates "*I fixed the code, it runs now,*" after seeing failed tests in-context, then its apparent beliefs (informed by the failed tests) contradict its statement. It is thus lying according to our definition. We operationalize beliefs following the approach of Ren et al. (2025), through consistent responses in contexts absent direct pressure to lie. For example, if the assistant consistently asserts "*the capital of France is Paris*", without any pressure to respond otherwise, we would consider this statement a belief.

Here are some excerpts from [Mahon 2016] which I felt were important to outlining the matter.

> L1 is the traditional definition of lying. According to L1, there are at least four necessary conditions for lying. First, lying requires that a person make a statement (statement condition). Second, lying requires that the person believe the statement to be false; that is, lying requires that the statement be untruthful (untruthfulness condition). Third, lying requires that the untruthful statement be made to another person (addressee condition). Fourth, lying requires that the person intend that that other person believe the untruthful statement to be true (intention to deceive the addressee condition).
>
> These four necessary conditions need to be explained before objections to L1 can be entertained and alternative definitions can be considered.

## 1.1 Statement Condition

According to the statement condition, lying requires that a person make a statement. Making a statement requires the use of conventional signs, or *symbols*. Conventional signs, such as "WOMEN" on the door to a restroom, are opposed to natural or causal signs, or *indices*, such as women coming in and out of a restroom, as well as signs that signify by resemblance, or *icons*, such as a figure with a triangular dress on the door to a restroom (cf. Grotius 2005, 2001; Pierce 1955; Grice 1989). Making a statement, therefore, requires the use of language. A commonly accepted definition of making a statement is the following: "$x$ states that $p$ to $y =_{\text{df}}$ (1) $x$ believes that there is an expression $E$ and a language $L$ such that one of the standard uses of $E$ in $L$ is that of expressing the proposition $p$; (2) $x$ utters $E$ with the intention of causing $y$ to believe that he, $x$, intended to utter $E$ in that standard use" (Chisholm and Feehan 1977, 150).

It is possible for a person to make a statement using American Sign Language, smoke signals, Morse code, semaphore flags, and so forth, as well as by making specific bodily gestures whose meanings have been established by convention (e.g., nodding one's head in response to a question). Hence, it is possible to lie by these means. If it is granted that a person is *not* making a statement when he wears a wig, gives a fake smile, affects a limp, and so forth, it follows that a person cannot be lying by doing these things (Siegler 1966, 128). If it is granted that a person is *not* making a statement when, for example, she wears a wedding ring when she is not married, or wears a police uniform when she is not a police officer, it follows that she cannot be lying by doing these things.

In the case of a person who does not utter a declarative sentence, but who curses, or makes an interjection or an exclamation, or issues a command or an exhortation, or asks a question, or says "Hello," then, if it is granted that she is *not* making a statement when she does any of these things, it follows that she cannot be lying by doing these things (Green 2001, 163–164; but see Leonard 1959).

An ironic statement, or a statement made as part of a joke, or a statement made by an actor while acting, or a statement made in a novel, is still a statement. More formally, the statement condition of L1 obeys the following three constraints (Stokke 2013a, 41):

i. If $x$ makes a statement, this does not entail that $x$ believes the statement to be true;
ii. If $x$ makes a statement, this does not entail that $x$ intends her audience to believe the statement to be true;
iii. If $x$ makes a statement, this does not entail that $x$ intends her audience to believe that $x$ believes the statement to be true.

The statement condition is to be distinguished from a different putative necessary condition for lying, namely, the condition that an *assertion* be made. The assertion condition is not a necessary condition for lying, according to L1. For example, if Yin, who does not have a girlfriend, but who wants people to believe that he has a girlfriend, makes the ironic statement "Yeah, right, I have a girlfriend" in response to a question from his friend, Bolin, who believes that Yin is secretly dating someone, with the intention that Bolin believe that he actually does have a girlfriend, then this 'irony lie' is a lie according to L1, although it is not an assertion.

## 2.4 Non-Deceptionism

Non-Deceptionists hold that an intention to deceive is not necessary for lying.

For Simple Non-Deceptionists (Augustine 1952 (cf. Griffiths 2003, 31); Aquinas 1952; Shibles 1985), there is nothing more to lying than making an untruthful statement. According to Aquinas, for example, a jocose lie is a lie. This position is not defended by contemporary philosophers.

For Complex Non-Deceptionists, untruthfulness is not sufficient for lying. In order to differentiate lying from telling jokes, being ironic, acting, etc., a further condition must be met. For some Complex Non-Deceptionists, that further condition is warranting the truth of the untruthful statement. For other Complex Non-Deceptionists, that condition is making an assertion.

Thomas Carson holds that it is possible to lie by making a false and untruthful statement to an addressee without intending to deceive the addressee, so long as the statement is made in a context such that one “warrants the truth” of the statement (and one does not believe oneself to be not warranting the truth of the statement), or one *intends* to warrant the truth of the statement:

(L12) A person *x* tells a lie to another person *y* iff (i) *x* makes a false statement *p* to *y*, (ii) *x* believes that *p* is false or probably false (or, alternatively, *x* does not believe that *p* is true), (iii) *x* states *p* in a context in which *x* thereby warrants the truth of *p* to *y*, and (iv) *x* does not take herself to be not warranting the truth of what she says to *y*. (Carson 2006, 298; 2010, 30)

(L13) A person *x* tells a lie to another person *y* iff (i) *x* makes a false statement *p* to *y*, (ii) *x* believes that *p* is false or probably false (or, alternatively, *x* does not believe that *p* is true), and (iii) *x* intends to warrant the truth of *p* to *y*. (Carson 2010, 37)

Carson includes the falsity condition in both of his definitions; however, he is prepared to modify both definitions so that the falsity condition is not required (Carson 2010, 39). He also holds that the untruthfulness condition is not stringent enough, since, if a speaker simply does “not believe” her statement to be true (but does not believe it to be false), or believes that her statement is “probably false” (but does not believe it to be false), then she is lying.

Carson gives two examples of non-deceptive lies: a guilty student who tells a college dean that he did not cheat on an examination, without intending that the dean believe him (since “he is really hard-boiled, he may take pleasure in thinking that the Dean knows he is guilty”), because he knows that the dean’s policy is not to punish a student for cheating unless the student admits to cheating, and a witness who provides untruthful (and false) testimony about a defendant, where there is a preponderance of evidence against the defendant, without the intention that the testimony be believed by anyone, in order to avoid suffering retaliation from the defendant and/or his henchmen (Carson 2006, 289; 2010, 21). Neither person is lying according to the definitions of lying of Simple Deceptionists (L1, L2, L3, L4, and L5) or Complex Deceptionists (L6, L7, L8, and L9) (cf. Simpson 1992, 631) or Moral Deceptionists (L10, L11). Both are lying according to L12 and L13, because each warrants the truth of his statement, even though neither intends to deceive his addressee.

It has been argued that the witness and the student do have an intention to deceive (Meibauer 2011, 282; 2014a, 105). It has also been argued that they are being deceptive, even if they lack an intention that their untruthful statements be believed to be true (Lackey 2013; but see Fallis 2015). However, it has also been argued that they fail to warrant the truth of their statements, and hence fail to be lying according to L12 and L13. One argument is that, in the witness example, the statement is coerced, and “Coerced speech acts are not genuinely assertoric” (Leland 2013, 3; cf. Kenyon 2010). “In the context of a threat of violent death, the mere fact that he is speaking under oath is not sufficient to institute an ordinary warranting context” (Leland 2013, 4). Another argument is that the

# Collaborators

**Ishaan Shrivastava** **visheshgupta14x@gmail.com** **praful.mohanan@gmail.com**
igarg2001@gmail.com

Correspondence:
ishaanshri95@gmail.com or @ishaanshri95 on discord